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CMS Website Generator

✓ Relevant source files

`hclient/widgets/cms/cmsTemplate.php``hclient/widgets/cms/editCMS2.js``hclient/widgets/cms/editCMS_ElementCfg.html``hclient/widgets/cms/editCMS_ElementCfg.js``hclient/widgets/cms/editCMS_SelectElement.html``hclient/widgets/cms/editCMS_SelectElement.js``hclient/widgets/cms/websiteRecord.php``hclient/widgets/cms/websiteScriptAndStyles.php`

The Heurist CMS Website Generator is a built-in feature that allows users to create and publish interactive websites directly from Heurist database content. This document explains the architecture, components, workflow, and functionality of the website generator system.

For information about the related Mapping System, see [Mapping System](#).

Purpose and Scope

The CMS Website Generator enables Heurist users to create websites without requiring extensive web development knowledge. Key capabilities include:

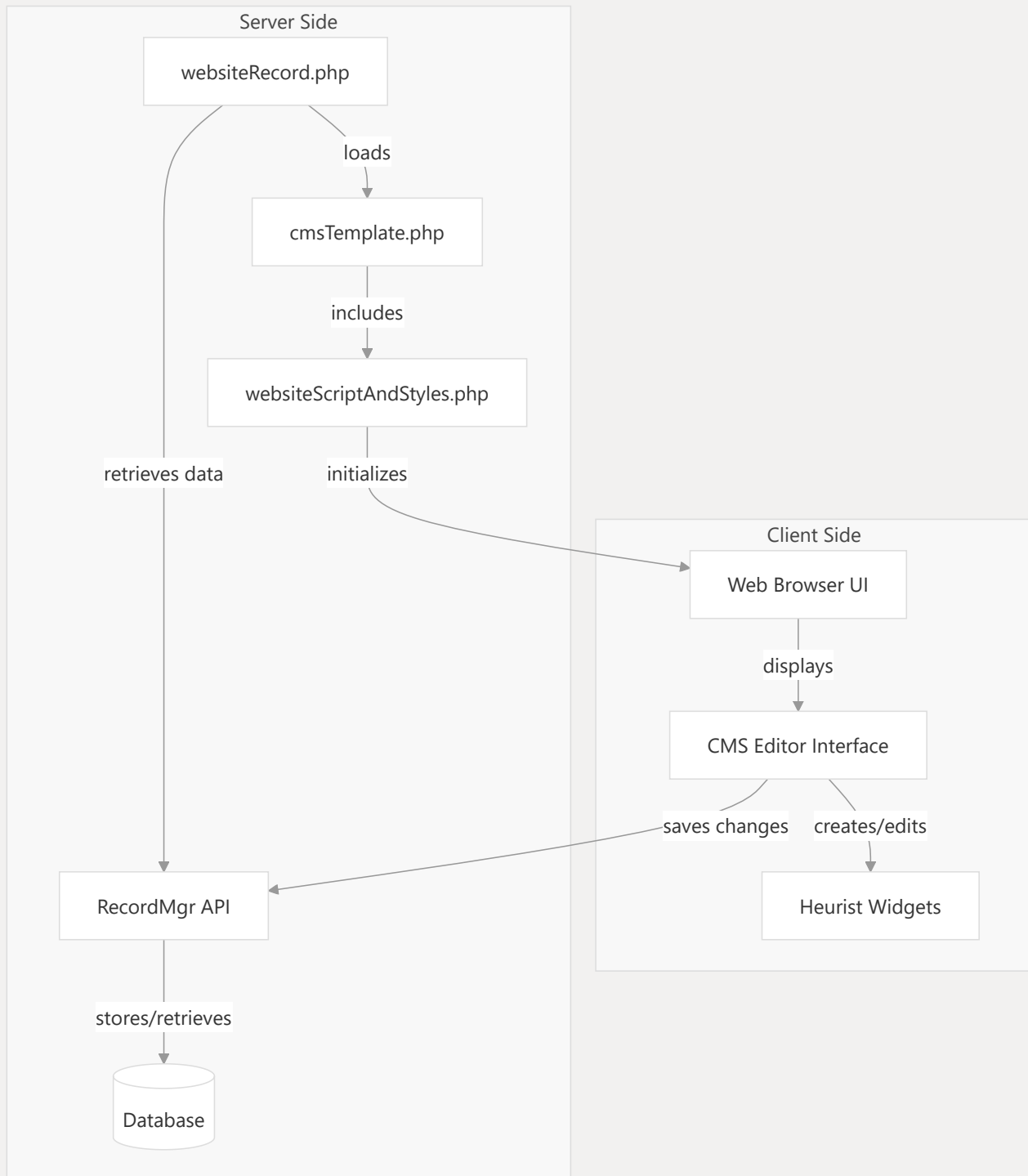
- Creating multi-page websites with navigation menus
- Designing page layouts with various content elements
- Embedding interactive Heurist widgets (search interfaces, maps, record lists)
- Customizing website appearance with templates and CSS
- Publishing content directly from the Heurist database

Sources:

- `hclient/widgets/cms/websiteRecord.php` 4-33
- `hclient/widgets/cms/cmsTemplate.php` 2-33

Architecture Overview

The CMS Website Generator consists of several interconnected components that handle different aspects of website creation, editing, and rendering:



The system follows a client-server model where:

- 1. Server-side components load website configuration from database records
- 2. Template files define the website structure
- 3. Client-side JavaScript enables editing and interaction
- 4. Database stores all website content and configuration

Sources:

- `hclient/widgets/cms/websiteRecord.php` 34-105
- `hclient/widgets/cms/websiteScriptAndStyles.php` 29-48
- `hclient/widgets/cms/editCMS2.js` 38-102

Core Components

Website Record (Entry Point)

The `websiteRecord.php` file serves as the entry point for website generation. It:

- Loads parameters from CMS Home or Web Page records
- Validates access permissions
- Prepares variables needed for the template (titles, images, styles)
- Loads the appropriate template file

This component translates database records into website content by extracting key fields:

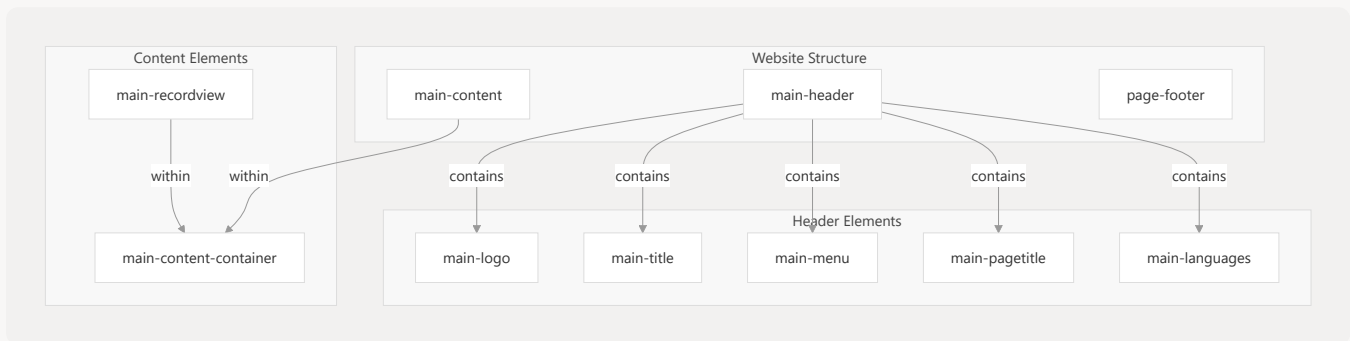
Field	Content	Used For
DT_NAME	Website title	Browser title, header text
DT_FILE_RESOURCE	Logo image	Website header
DT_CMS_CSS	Custom CSS	Website styling
DT_CMS_SCRIPT	Custom JavaScript	Website interactivity
DT_CMS_TEMPLATE	Template name	Website structure
DT_EXTENDED_DESCRIPTION	HTML content	Page content

Sources:

- `hclient/widgets/cms/websiteRecord.php` 107-146
- `hclient/widgets/cms/websiteRecord.php` 231-321

Website Template

The `cmsTemplate.php` defines the default HTML structure that contains required elements:



The template includes:

- Essential elements (`main-header` , `main-menu` , `main-content`)
- Optional elements (`main-logo` , `main-pagetitle` , `main-languages`)
- Default styling for all components
- Placeholders for dynamic content

Sources:

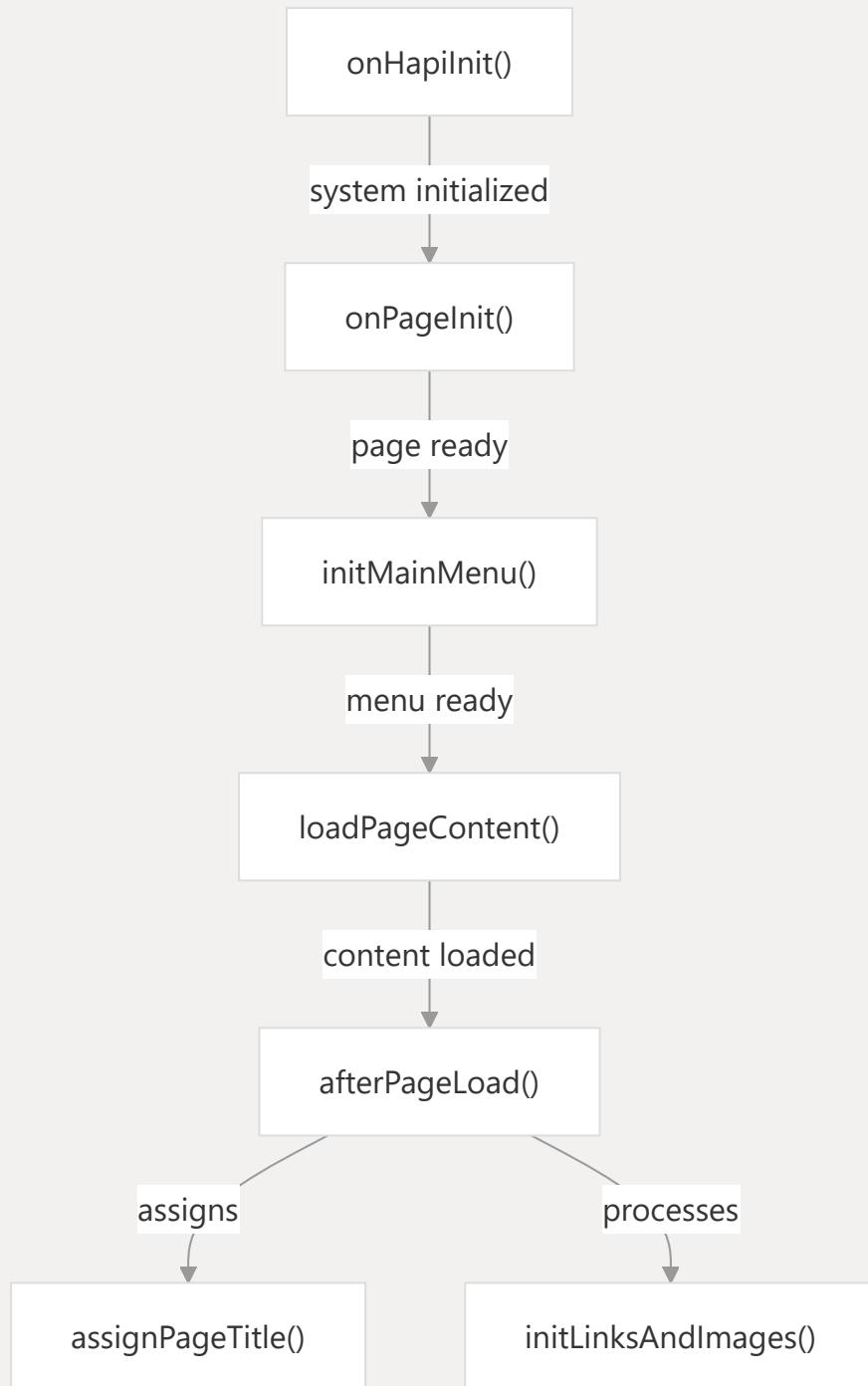
- `hclient/widgets/cms/cmsTemplate.php` 71-197
- `hclient/widgets/cms/cmsTemplate.php` 202-265

Scripts and Styles Manager

The `websiteScriptAndStyles.php` file serves as the bridge between the server-side template and client-side functionality:

- Loads required JavaScript libraries and CSS
- Defines global variables for website configuration
- Implements core functions for page initialization and content loading
- Handles navigation between pages and content rendering

Key functions include:



These functions create the workflow for page loading and initialization.

Sources:

- `hclient/widgets/cms/websiteScriptAndStyles.php` 49-144
- `hclient/widgets/cms/websiteScriptAndStyles.php` 303-407
- `hclient/widgets/cms/websiteScriptAndStyles.php` 415-440

CMS Editor Interface

The `editCMS2.js` creates the editor interface for website content manipulation:

- Provides a side panel with website and page structure views
- Enables adding, removing, and configuring page elements
- Manages element selection and property editing
- Handles saving changes back to the database

The editor interface is activated when a user clicks the "website editor" button and provides two main views:

1. Site view - for managing the website structure (pages and navigation)
2. Page view - for designing the content of individual pages

Sources:

- `hclient/widgets/cms/editCMS2.js` 43-103
- `hclient/widgets/cms/editCMS2.js` 106-316

Element Configuration

The `editCMS_ElementCfg.js` provides the interface for configuring element properties:

- Basic properties (name, ID, classes)
- Layout properties (display type, flex settings, dimensions)
- Appearance properties (border, background, margins, padding)
- Widget-specific properties for interactive elements

This component creates a form interface with multiple sections for different property categories, allowing fine-grained control over each element's appearance and behavior.

Sources:

- `hclient/widgets/cms/editCMS_ElementCfg.js` 26-96
- `hclient/widgets/cms/editCMS_ElementCfg.html` 21-406

Element Selection

The `editCMS_SelectElement.js` creates a dialog for selecting elements to add to a page:

- Categorizes elements by type (layouts, content, widgets)
- Provides descriptions and usage information
- Enables filtering and searching for specific element types
- Handles insertion of selected elements into the page

The selection dialog organizes elements into logical groups:

- Layout elements (Group, Accordion, Tabs, Cardinal)
- Content elements (Text box, Columns, Text on banner)
- Widget elements (Search, Results, Maps, Navigation)

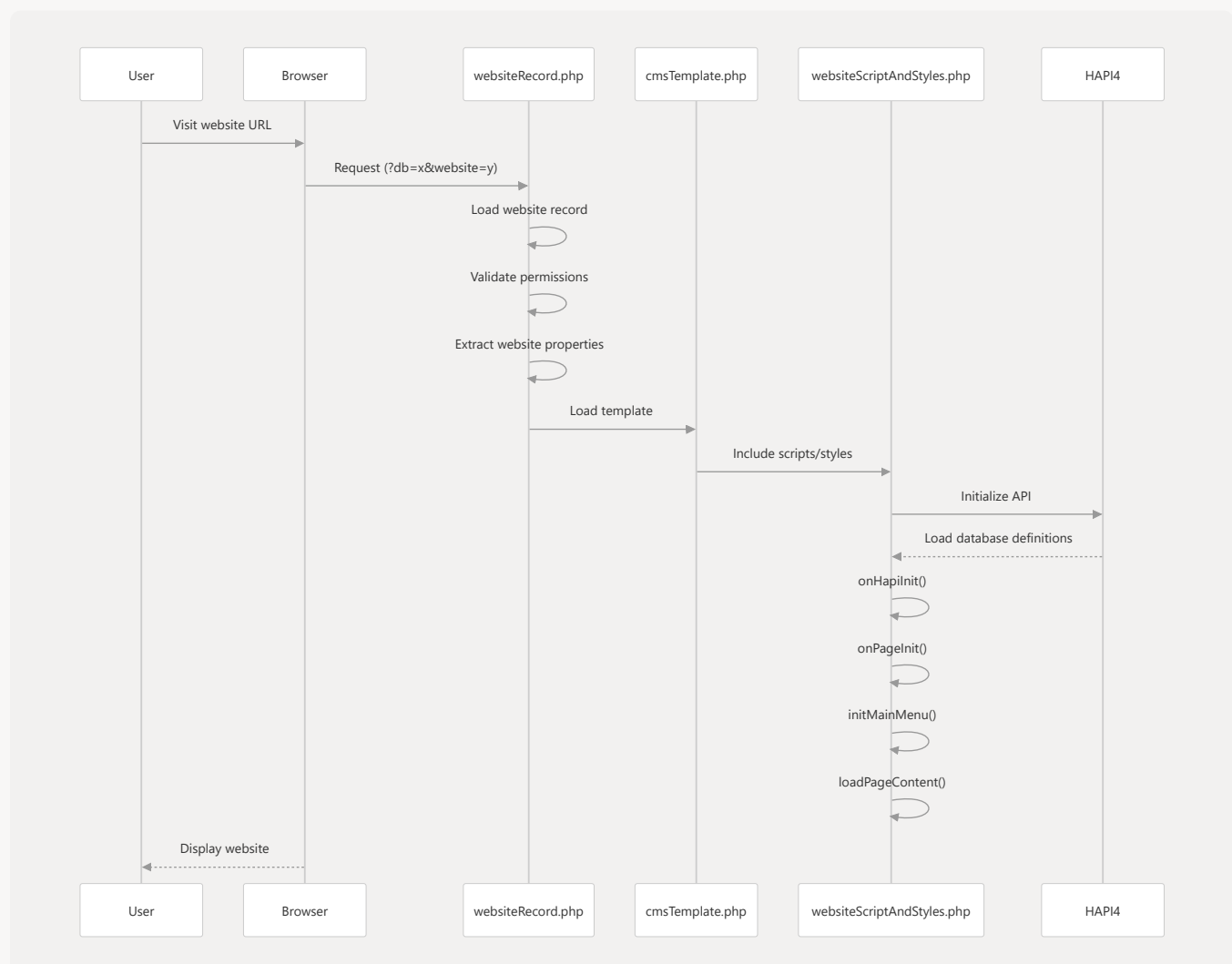
Sources:

- `hclient/widgets/cms/editCMS_SelectElement.js` 22-186
- `hclient/widgets/cms/editCMS_SelectElement.html` 46-94

Website Generation Workflow

The CMS Website Generator follows a defined workflow from initialization to interaction:

Initialization Process



The initialization process turns database records into an interactive website by:

1. Loading the website configuration from a CMS Home record

2. Applying the template structure (default or custom)
3. Initializing JavaScript components for interactivity
4. Loading the initial page content

Sources:

- `hclient/widgets/cms/websiteRecord.php` 35-54
- `hclient/widgets/cms/websiteScriptAndStyles.php` 304-408

Page Loading Process

When navigating between pages, the system:

1. Retrieves page content from cache or database
2. Updates the page title and metadata
3. Loads content into the `main-content` container
4. Initializes Heurist widgets within the content
5. Applies page-specific CSS and JavaScript

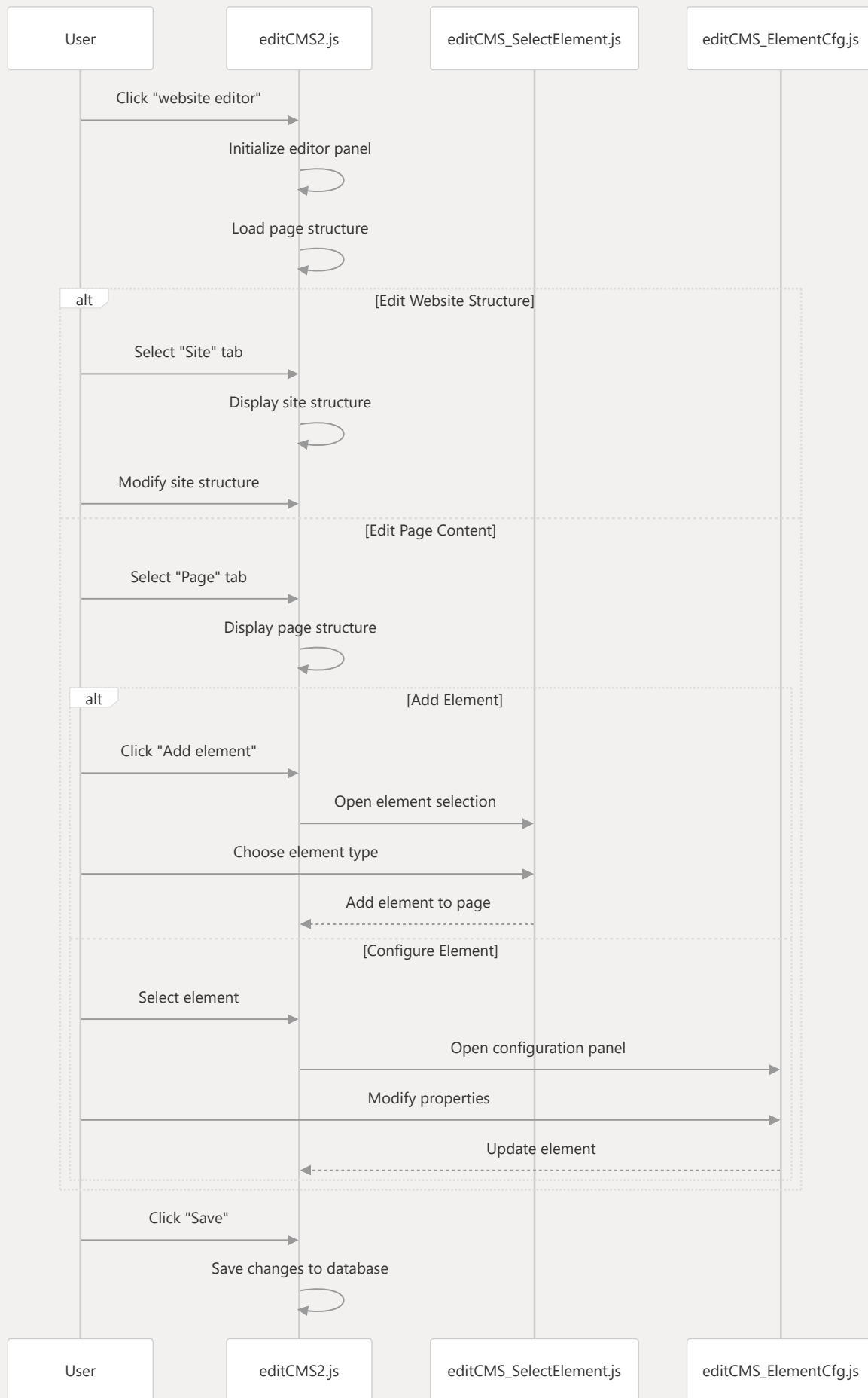
This process enables navigation without full page reloads, creating a smoother user experience.

Sources:

- `hclient/widgets/cms/websiteScriptAndStyles.php` 479-657
- `hclient/widgets/cms/websiteScriptAndStyles.php` 751-866

Content Editing Process

The content editing workflow allows website administrators to modify pages:



This process allows complete customization of both the website structure and individual page content.

Sources:

- `hclient/widgets/cms/editCMS2.js` 90-285
- `hclient/widgets/cms/editCMS2.js` 446-510

Element Types and Configuration

The CMS Website Generator supports various element types that can be added to pages:

Layout Elements

Layout elements help organize page content:

Element	Description	Configuration Options
Group	Container for other elements	Display type, flex settings, dimensions
Accordion	Collapsible sections	Initial state, header appearance, animation
Tabs	Tabbed interface	Tab labels, orientation, appearance
Cardinal	North/South/East/West/Center layout	Pane sizes, initial states, resizable

Sources:

- `hclient/widgets/cms/editCMS_SelectElement.js` 42-47
- `hclient/widgets/cms/editCMS_ElementCfg.html` 38-97
- `hclient/widgets/cms/editCMS_ElementCfg.html` 150-243

Content Elements

Content elements display static or editable content:

Element	Description	Configuration Options
Simple text box	WYSIWYG text editor	Formatting, media embedding, styles
Text in 2 columns	Two-column layout	Column widths, spacing, content
Text in 3 columns	Three-column layout	Column widths, spacing, content
Text on banner	Text over background image	Background image, text position, overlay

Sources:

- `hclient/widgets/cms/editCMS_SelectElement.js` 51-56
- `hclient/widgets/cms/editCMS_ElementCfg.html` 112-146

Widget Elements

Widget elements provide dynamic content and interactivity:

Widget	Description	Configuration Options
Filter	Search input field	Default search, appearance, behavior
Saved filters	Tree of saved filters	Default filters, appearance, behavior
Result list	Search results display	View mode, columns, sorting
Map	Geographic visualization	Default view, layers, controls
Timeline	Chronological visualization	Scale, grouping, appearance
Network graph	Relationship visualization	Layout, node styles, edge styles
Menu	Navigation menu	Orientation, depth, appearance

Sources:

- `hclient/widgets/cms/editCMS_SelectElement.js` 58-73
- `hclient/widgets/cms/editCMS2.js` 579-604

Common Tasks and Examples

Creating a New Website

To create a new website:

1. Create a CMS Home record (RT_CMS_HOME) in the database
2. Set basic properties (title, logo, banner)
3. Configure optional settings (template, custom CSS/JS)
4. Make the record publicly accessible to publish the website

The website will be accessible at `?db=database&website=record_id`.

Adding Pages to a Website

Pages can be added to a website:

1. In the CMS editor, select the "Site" tab
2. Click the "+" button to add a new page
3. Enter a title for the page
4. The page will appear in the navigation menu
5. Select the page to edit its content

Designing Page Content

Page content is designed using the CMS editor:

1. Select the "Page" tab in the editor
2. Add layout elements (Group, Accordion, Tabs) to organize content
3. Add content elements (Text, Columns) to display static content
4. Add widget elements (Search, Results, Maps) for interactive features
5. Configure each element's properties through the properties panel

Customizing Website Appearance

Website appearance can be customized at multiple levels:

1. Global styling through the CMS Home record's CSS field
2. Page-specific styling through individual page CSS fields
3. Element-specific styling through the properties panel
4. Custom templates for complete control over the website structure